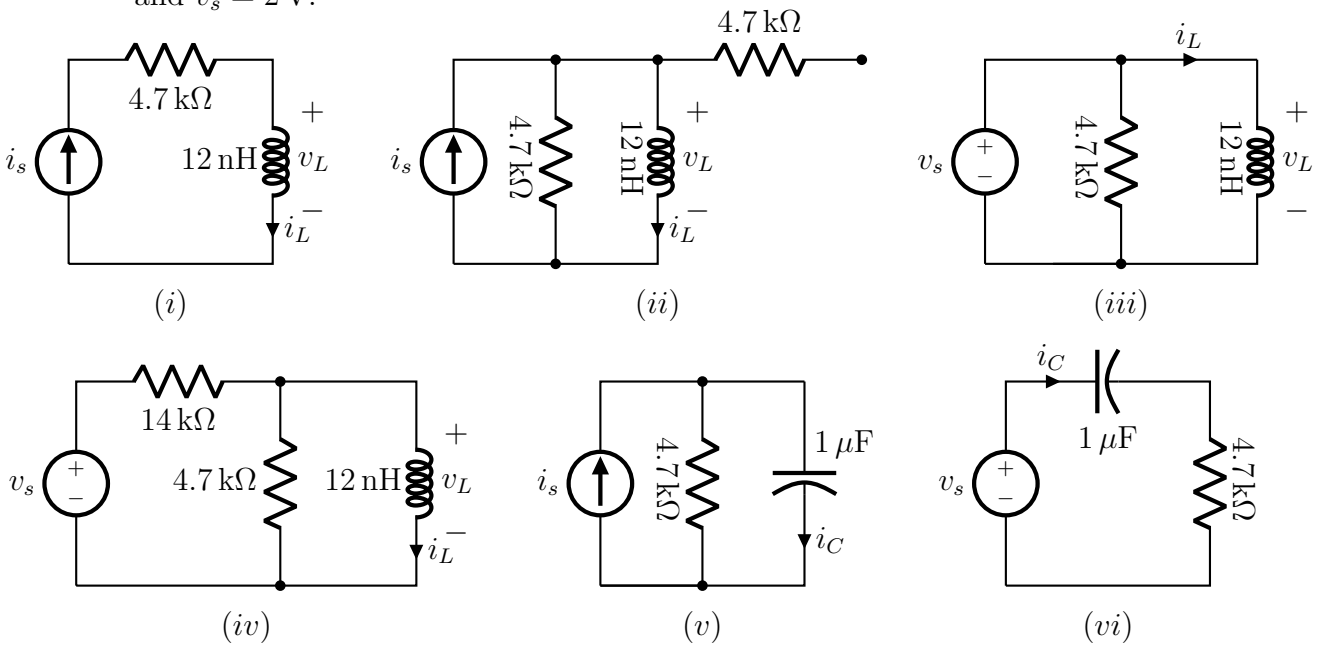
EE2015: Electric Circuits and Networks

Tutorial 1

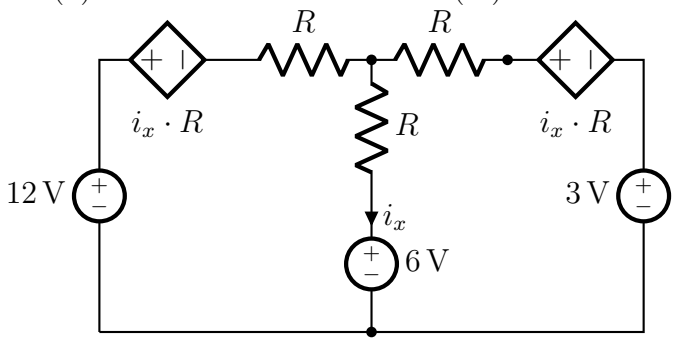
(August 8, 2025)

1. (a) For the circuits given in (i) to (iv), calculate v_L and i_L. Assume $i_s = 1\,\text{mA}$ and $v_s = 2\,\text{V}$.

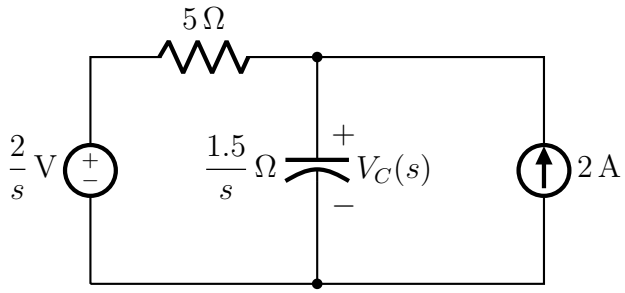
   (b) For the circuits given in (v) and (vi), find out the currents i_C. Assume $i_s = 1\,\text{mA}$ and $v_s = 2\,\text{V}$.



2. In the circuit shown on the right, determine the power *delivered* by the 6 V source when $R = 10\,\text{k}\Omega$.



3. Consider the circuit shown on the left.

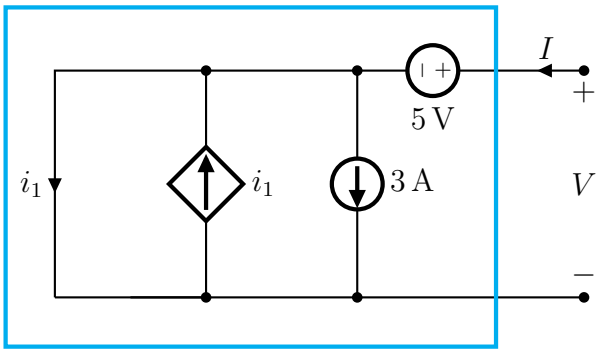


   (a) Calculate $V_C(s)$.

   (b) Determine $v_C(t)$ for $t \geq 0$.

   (c) Draw the time-domain representation of the circuit.

4.

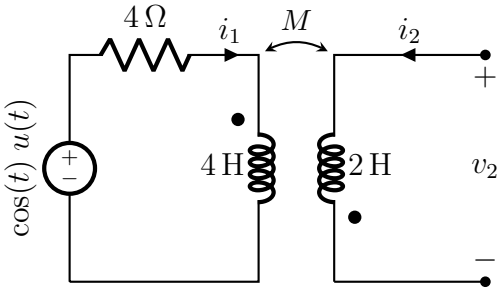


Draw the I–V characteristics of the circuit shown on the left.

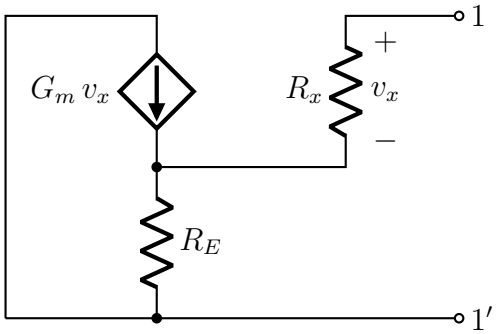
5. Consider the circuit shown on the right, where the mutual inductance $M = 0.4\,\mathrm{H}$.

   (a) Write the differential equation for $i_1(t)$. Assuming zero initial conditions, solve the differential equation using the Laplace transform approach.

   (b) Find $v_2(t)$.



6.



Consider the circuit show on the left. What is the equivalent resistance as seen at the terminal pair $(1, 1')$?

$G_m = 1\,\mathrm{mS}$
$R_x = 10\,\mathrm{k\Omega}$
$R_E = 20\,\mathrm{k\Omega}$

Hint: Connect a 1 A current source across terminals $(1, 1')$ and calculate $V_{11'}$.